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Class balancing diversity multimodal ensemble for Alzheimer's disease diagnosis and early detection

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ABSTRACT

Alzheimer's disease (AD) poses significant global health challenges due to its increasing prevalence and associated societal costs. Early detection and diagnosis of AD are critical for delaying progression and improving patient outcomes. Traditional diagnostic methods and single-modality data often fall short in identifying early-stage AD and distinguishing it from Mild Cognitive Impairment (MCI). This study addresses these challenges by introducing a novel approach: multImodal enseMble via class BALancing diversity for iMbalancEd Data (IMBALMED). IMBALMED integrates multimodal data from the Alzheimer's Disease Neuroimaging Initiative database, including clinical assessments, neuroimaging phenotypes, biospecimen, and subject characteristics data. It employs a new ensemble of model classifiers, designed specifically for this framework, which combines eight distinct families of learning paradigms trained with diverse class balancing techniques to overcome class imbalance and enhance model accuracy. We evaluate IMBALMED on two diagnostic tasks (binary and ternary classification) and four binary early detection tasks (at 12, 24, 36, and 48 months), comparing its performance with state-of-the-art algorithms and an unbalanced dataset method. To further validate the proposed model and ensure genuine generalization to real-world scenarios, we conducted an external validation experiment using data from the most recent phase of the ADNI dataset. IMBALMED demonstrates superior diagnostic accuracy and predictive performance in both binary and ternary classification tasks, significantly improving early detection of MCI at a 48-month time point and showing excellent generalizability in the 12-month task during external validation. The method shows improved classification performance and robustness, offering a promising solution for early detection and management of AD.

1. Introduction

Alzheimer's disease (AD) is a neurodegenerative disorder characterized by progressive cognitive deterioration, along with declining activities of daily living and behavioral changes. It primarily affects brain regions involved in memory, cognition, and language, leading to significant impairment in an individual's ability to engage in conversation and interact with their surroundings (It and Know, 1986). AD is the most common form of dementia, affecting over 55 million people worldwide in 2023, with 10 million new cases per year (World Health Organization, 2023) implying a new case diagnosed every three seconds (World Health Organization et al., 2018). The World Health Organization estimates that this number will increase to 78 million people by 2030 and to 139 million by 2050 (Shin, 2022). In Europe, the social and economic costs of dementia were about €238.6 billion in

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2010 and, based on the expected increase in people with dementia, it is estimated to increase to €343 billion by 2050 (Maresova et al., 2020). Considering the escalating global prevalence of AD and the potential economic and societal burdens it entails, there is a pressing need for innovative and effective approaches to address both the diagnosis and early detection of AD.

Current available treatments decelerate only the progression of AD and no treatment developed so far is capable of curing a patient with this disease. Thus, developing strategies for the detection of AD at early stages, before the manifestation of clinical symptoms, is crucial for timely treatment and progression delay (Alhazmi and Albratty, 2022). Generally, patients can be categorized into three groups at diagnosis: AD, Mild Cognitive Impairment (MCI), and Cognitively Normal (CN). MCI serves as an intermediate stage between CN and AD, defined to describe individuals experiencing mild cognitive deficits yet still capable of performing daily activities. MCI is further subdivided into Early Mild Cognitive Impairment (EMCI) and Late Mild Cognitive Impairment (LMCI), reflecting different degrees of cognitive decline: EMCI represents milder symptoms, while LMCI involves more severe symptoms similar to early-stage AD. Patients with MCI face a higher risk of progressing to Alzheimer's disease, highlighting the importance of accurately diagnosing MCI and predicting its conversion to AD at an early stage (Alzheimer's Association, 2019; Iddi et al., 2019). In this context, advancements in artificial intelligence (AI) are increasingly pivotal in enhancing the early detection of MCI (Vrahatis et al., 2023; Subasi, 2020). These technologies offer the potential to improve diagnostic accuracy and predict disease progression more effectively. However, the application of AI in this field is not without challenges. In particular, class imbalance is one of the most relevant challenges, where the number of patients who convert from MCI to AD is significantly lower than those who do not (Dubey et al., 2014). This imbalance can lead to biased machine learning (ML) models that favor the majority class, typically CN, resulting in poor identification of potential AD or MCI cases. Furthermore, to date, the AD diagnosis relies primarily on a single data source, also known as modality, such as neuropsychological tests, which assess cognitive abilities like memory, attention, and language that decline with disease progression. However, using a single modality often fails to provide the comprehensive information needed to identify distinguishing patterns for early AD diagnosis, making it challenging for clinicians to detect the disease at its initial stages (Sharma and Mandal, 2022). In this context, the integration of multiple data sources offers the potential for a more comprehensive and accurate assessment of an individual's risk of developing AD (Venugopalan et al., 2021). In addition, biomarkers from various modalities, which are measurable indicators of the severity or presence of some disease state, play a prominent role in the diagnosis of AD. Multiple biomarkers have been explored and investigated through brain structural, neurochemical, and behavioral studies, offering diverse information (Mandal et al., 2015, 2012). Key biomarkers include brain imaging markers such as Magnetic Resonance Imaging (MRI) and Positron Emission Tomography (PET), Cerebrospinal Fluid (CSF), and genetic markers, such as those in the ApoE gene, which is involved in cholesterol transport and is linked to an increased risk of AD (Namboori et al., 2011). Recognizing the diverse nature of these biomarkers, the integration of multiple modalities offers a promising avenue in the diagnosis and early detection of AD, providing a comprehensive understanding and enhancing AI model accuracy.

This study is driven by a twofold motivation: clinical and methodological. From a clinical perspective, AD continues to present unresolved challenges in both diagnosis and prognosis. The growing incidence of AD, coupled with its societal and economic burden, underscores the critical need for reliable tools that enable early detection and effective monitoring of disease progression. Early diagnosis, in particular, offers the potential to slow disease progression, improve quality of life, and reduce healthcare costs. However, there remain open questions in accurately identifying MCI and predicting its conversion to AD, especially at its earliest stages when clinical interventions may be most effective. From a methodological standpoint, the challenges posed by diverse data modalities and class imbalance introduce biases that compromise the performance of AI models, highlighting the need for innovative approaches. To address these challenges, we propose a novel imbalance method: multImodal enseMble via class BALancing diversity for iMbalancEd Data (IMBALMED). IMBALMED employs an ensemble of model classifiers, each trained with different class balancing techniques, and leverages the integration of multimodal tabular data from the Alzheimer's Disease Neuroimaging Initiative (ADNI) (Anon, 2024) database. To achieve this goal, we provide the following contributions:

- Performance Evaluation: we assess IMBALMED's effectiveness in diagnosing AD at the recruitment start, distinguishing between CN and AD (binary classification) as well as CN, AD, and MCI (ternary classification). Additionally, we predict disease progression with binary tasks (AD vs. CN+MCI) at 12, 24, 36, and 48 months, addressing various stages of AD's progression.
- Integration of multimodal data: we harness the complementary information provided by diverse sources, by combining data across four different modalities, i.e., clinical assessments, biospecimen data, subject characteristics, and neuroimaging phenotypes, all obtained from the ADNI database.
- Class balancing diversity: we adjust for the diversity in data representation by training an ensemble of models on different class distributions for each modality fostering a comprehensive learning process that captures diverse data characteristics.
- Robust comparative analysis in different scenarios: we ensure the robustness and reliability of our methodology, by comparing our approach with an unbalanced dataset approach and nine state-of-the-art algorithms specifically designed to handle imbalanced data. Additionally, we carry out statistical tests to verify that our results are significantly better than our competitors. Furthermore, we perform an external validation using data from the most recent phase of the ADNI dataset to assess the generalization capabilities of our model in real-world scenarios, ensuring that its performance is not limited to the primary dataset.

The rest of this manuscript is organized as follows: Section 2 provides background on the field of AD, beginning with key challenges and followed by a review of state-of-the-art studies. Section 3 introduces the datasets used in this work and the inclusion criteria for selected ADNI patients, whereas Section 4 presents our method. Section 5 describes the experimental configuration; Section 6 reports and discusses the results. Finally, Section 7 provides concluding remarks.

2. Background

AI is revolutionizing medical diagnostics, particularly in the detection of AD (Sharma and Mandal, 2022; Venugopalan et al., 2021). These technologies automate the analysis of complex data, enabling ML models to extract and interpret critical insights, essential for creating effective predictive tools. In this section, we will briefly discuss the primary challenges in the field of AD: the use of multiple data modalities to enhance AI model performance and the class imbalance in datasets, where examples are unevenly distributed across different classes. We will then review some state-of-the-art studies selected based on the following criteria: use of multiple data modalities, methods for addressing the dataset imbalance problem, and data obtained from the ADNI database. This ensures consistency with our selected database and enables meaningful result comparisons. ADNI is the most comprehensive public dataset on Alzheimer's Disease, offering the broadest range of longitudinal tabular data. Additionally, its widespread use in scientific literature facilitates easier comparison of results.

AD's diagnosis with AI can be traced back to the utilization of individual data modalities, commonly known as unimodal approaches. Some of the key unimodal data sources used for AD diagnosis are: MRI, PET, CSF, genetic information, and demographic information (Weiner

et al., 2017). Whilst unimodal approaches have significantly advanced our knowledge of AD, it is increasingly recognized that a multimodal approach, which integrates multiple modalities, is essential for capturing the full complexity of the disease (Di Teodoro et al., 2025; Guarrasi et al., 2024a,b; Guarrasi and Soda, 2023; Venugopalan et al., 2021; Caruso et al., 2022; Tortora et al., 2023). This method, denoted as multimodal fusion, has been shown to surpass the performance of unimodal models in detecting early cognitive impairment and predicting AD conversion (Venugopalan et al., 2021). Multimodal fusion can occur at various levels, including late, intermediate, and early fusion. Late fusion involves combining the outputs of individual unimodal models, integrating their decisions at a final stage; intermediate fusion combines features at an intermediate layer within the network; early fusion, on the other hand, integrates raw data from different modalities right at the input level. Late fusion offers a significant advantage over others by ensuring a consistent representation of decisions across all modalities and simplifies the combination of multiple modalities at the decision level, making it easier to employ the most optimal analysis methods for each modality independently (Sharma and Mandal, 2022). Researchers are continually exploring various fusion methods, seeking the optimal approach to combine diverse data sources for improved classifier performance.

Another ongoing challenge is training AI models with imbalanced datasets (Johnson and Khoshgoftaar, 2019). Models trained on such datasets may achieve high accuracy by predominantly predicting the majority class, but they often struggle to accurately identify the minority class cases. Several techniques are used to address the data imbalance issue; three primary approaches are: undersampling, oversampling, and ensemble learning. Undersampling involves randomly removing data points from classes that have an excess of samples to balance class distributions. Whilst it promotes equal class sizes and faster training times, under-sampling risks discarding potentially informative data. Oversampling, in contrast, generates new synthetic data from under-represented classes to equalize class sizes, though it risks introducing overfitting (Brownlee, 2020). Several studies have shown that undersampled datasets yield stable and promising results, compared to oversampled datasets (Dubey et al., 2014). Finally, ensemble learning leverages multiple models to achieve better predictive performance than what could be obtained from any of the models alone (Mogensen et al., 2025; Ruffini et al., 2024; Guarrasi and Soda, 2022). This method has been shown to improve model performance (Fort et al., 2019), outperforming single-model approaches in handling imbalanced data (Feng et al., 2018), despite being less explored (Galar et al., 2011; Aguiar et al., 2023).

To the best of our knowledge, there are only a few contributions in the literature that address both class imbalance and multimodal learning on the ADNI dataset to improve Alzheimer's disease diagnosis (Sun et al., 2021; Feng et al., 2024; Velazquez and Lee, 2022; Brand et al., 2020). Sun et al. (2021) proposed a new algorithm combining SMOTE (Chawla et al., 2002) with a multi-dimensional Gaussian probability density hypothesis (Naseriparsa et al., 2020), utilizing a Random Forest (RF) classifier, but without explicitly detailing the specific validation method employed. Their method integrated data from six modalities, including MRI and PET images, genetics, cognitive tests, CSF, and blood biomarkers at the feature level. This approach achieved an accuracy of 95.62% and a recall of 90.36% on CN vs. AD classification. Additional results showed an accuracy of 90.82% and a recall of 80.25% for EMCI vs. AD, and 85.15% accuracy with 68.12% recall for LMCI vs. AD. Feng et al. (2024) proposes a scalable framework for classifying CN vs. MCI vs. AD, using eight different modalities from the ADNI dataset. These modalities include structural and functional imaging (MRI, and FDG-PET) as well as cognitive and clinical assessments, such as Montreal Cognitive Assessment, and Neuropsychiatric Inventory Questionnaire. The method employs a tunable modality-agnostic Transformer architecture, which reduces modality-specific parameters to mitigate overfitting and instability. Additionally, task-specific loss functions, such as alignment loss and N-pair loss, were integrated to address the class imbalance by improving robustness and preventing early overfitting. The framework also introduces a decoupled multi-modal alignment process, separating representation-level and task-level alignment to improve feature exploration and ensure resilience to varying modality combinations. Evaluated using an 8:2 patient-level split for training and testing, the model achieved an accuracy and weighted F1 score of 73%, with performance improving as more modalities were integrated. Velazquez and Lee (2022) developed a multimodal ensemble model to predict AD conversion in EMCI patients. The model leverages two key data sources: Diffusion Tensor Imaging (DTI), which is a type of MRI technique that specifically measures the diffusion of water molecules in brain tissue, and Electronic Health Records (EHR), including patient demographics, genetic biomarkers, and neuropsychological test scores. The model combines a Convolutional Neural Network to process the DTI images with an RF classifier for EHR-based data, employing a cross-validation method. A grid search was employed to optimize the weighting between the two classifiers, improving the overall prediction accuracy. Given the inherent class imbalance (fewer patients who convert to AD), oversampling and data augmentation techniques were applied to address the issue, resulting in a 98.81% accuracy in predicting the conversion from EMCI to AD. Brand et al. (2020) proposed the Task Balanced Multimodal Feature Selection (TBMFS) method, which balances feature selection across genetic and neuroimaging data for a binary classification task to distinguish between AD and non-AD (CN and MCI patients). The TBMFS method balances feature selection by jointly optimizing the feature selection process and the classification task through a combined objective function, employing six-fold cross-validation. By integrating matrix factorization and support vector machines (SVM), the TBMFS method achieved a balanced accuracy of 72.8%, outperforming other models like SVM (59.8%) and logistic regression (62.9%).

As aforementioned, to the best of our knowledge, (Sun et al., 2021; Feng et al., 2024; Velazquez and Lee, 2022; Brand et al., 2020) are the only studies addressing the issue of imbalanced datasets and using multimodal data from the ADNI database. This highlights a gap in the AD's field utilizing the ADNI dataset. Most existing balancing methods focus on creating balanced subsets using all samples from the minority class and varying portions of the majority class (Aguiar et al., 2023). In contrast, our study aims to explore all possible combinations of minority and majority class samples, under the hypothesis that the diversity of the resulting subsets may provide more informative data for the model and thereby improve performance. Additionally, no literature presented before also evaluates AI models in early detection scenarios, despite the importance of early intervention for improving patient outcomes and slowing disease progression. Furthermore, many studies in the field of AD often rely on DL methods, which, despite their effectiveness, come with a substantial computational cost compared to traditional ML approaches (Schwartz et al., 2020). In this study, therefore, we focus on utilizing tabular data, including features extracted from images, and integrating it with ML classifiers, as ML methods have been shown to perform better with this type of data (Shwartz-Ziv and Armon, 2022).

3. Materials

3.1. Dataset and pre-processing

Data for this study were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (Anon, 2024), a cornerstone in AD's research, due to its extensive cohort and diverse range of data modalities. The dataset encompasses a wide spectrum of patient groups, from CN individuals to those diagnosed with MCI and AD, making it particularly suited for multimodal analyses. ADNI consists of five phases: ADNI 1 (2004–2010), GO (2009–2011), 2 (2011–2016), 3 (2016–2022), and 4 (2022-present). For this study, we used data

Table 1
Summary of selected ADNI data modalities measured at baseline, including the number of features and percentage of missing data for diagnostic and early detection tasks.

Modality	# of features		% of missing data	
	Diagnostic tasks	Early detection tasks	Diagnostic tasks	Early detection tasks
Assessment	37	110	31.06	30.54
Biospecimen	1	4	4.48	16.82
ImageAnalysis	9	10	15.34	11.29
SubjectCharacteristics	6	7	1.22	0.18

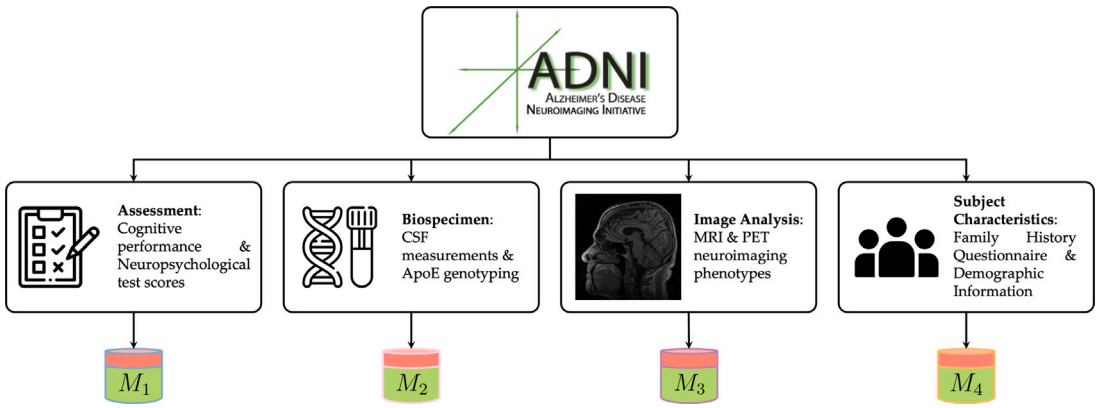


Fig. 1. Overview of selected data modalities from the ADNI dataset. Each modality is depicted in a labeled box, accompanied by an icon representing the data type and specific tests included in the modality. Arrows indicate the flow of data from the ADNI database to its integration into our final dataset. For notation, see Section 4.1.

from the first four phases of ADNI, as the most recent phase (ADNI 4) lacked key data modalities necessary for our analysis at the time of data acquisition (October 2023). Each phase is characterized by specific goals: ADNI 1 aimed to develop biomarkers for clinical trials; ADNI GO investigated biomarkers in early disease stages; ADNI 2 built on this by refining biomarkers as predictors of cognitive decline and outcome measures; and ADNI 3 focused on using functional imaging techniques in clinical trials.

We utilized four data modalities from the ADNI dataset collected at the start of the recruitment, referred to as the baseline: Assessment, Biospecimen, Image analysis, and Subject Characteristics. A visual representation of these selected modalities is shown in Fig. 1, while Table 1 provides a summary of the number of features and the percentage of missing data for each modality. Furthermore, a detailed list of selected features for each modality and task is provided in Table A.11 in Appendix A. The Assessment modality includes cognitive performance measures from the Functional Activities Questionnaire, Geriatric Depression Scale, and Modified Hachinski Ischemia Scale, along with neuropsychological test scores such as the Alzheimer's Disease Assessment Scale, Clinical Dementia Rating, (Modified) Preclinical Alzheimer's Cognitive Composite, and Mini-Mental State Examination. The Biospecimen modality includes CSF measurements (e.g., amyloid-beta, tau, phosphorylated tau), ApoE genotyping, and laboratory data (e.g., blood, urine, and chemistry panels). Genetic data like Genome-Wide Association Studies and Whole Genome Sequencing were excluded due to format inconsistencies with our tabular data needs. Image analysis, derived from MRI and PET scans, provides structural and functional brain data, including key neuroimaging biomarkers like hippocampal volume, entorhinal cortical thickness, and glucose metabolism in specific brain regions. These neuroimaging biomarkers are critical for understanding the structural and metabolic changes in the brain associated with AD. The Subject Characteristics modality encompasses demographic information (e.g., age, gender, education, weight, and birth date) and family history data derived from the Family History Questionnaire, which provides insight into genetic risk factors for Alzheimer's disease by detailing the parental and sibling history of AD.

The preprocessing pipeline was designed to ensure the quality, consistency, and interpretability of the data while addressing common challenges such as missing data and variability in scale across modalities. Initially, features with more than 50% missing data were excluded from the analysis. This threshold was chosen based on a balance between retaining informative features and minimizing potential bias introduced by excessive imputation. Studies investigating datasets with increasing proportions of missing data commonly examine a maximum of 50% missingness, demonstrating that higher levels of missingness are associated with increased variability in effect estimates (Madley-Dowd et al., 2019; Mishra and Khare, 2014). Additionally, scenarios with more than 50% missingness are often limited in their applicability to larger studies due to reliance on small sample sizes (e.g., less than 200 samples) or the absence of auxiliary variables to support imputation (Madley-Dowd et al., 2019; Lee and Huber Jr., 2011). These findings support the practical decision to exclude features exceeding 50% missingness as a means of preserving the robustness and reliability of the analysis.

To prepare the data for modeling, we applied One-Hot Encoding to categorical variables and normalized the data to ensure uniform scale across different modalities, a critical step to prevent features with larger ranges from dominating the model and to improve the convergence of optimization algorithms. Missing values were imputed using the k-NN algorithm with k set to 5. Normalization and imputation steps were carefully differentiated between the training set and the validation/test sets. For the training set, we calculated normalization parameters, i.e., mean and standard deviation, and imputed missing values based on its data. For the validation and test sets, we used the normalization parameters derived from the training set's pre-processing and applied the imputation method fitted on the train set to ensure consistency across all subsets and avoid introducing bias.

3.2. Patient selection

Our analysis involved patients from the ADNI study and we chose patients from the ADNI 1, GO, 2, and 3 phases. For the diagnostic tasks, we differentiated between CN, AD, and MCI through binary (CN vs.

Table 2
Distribution of patients for the diagnostic and early detection tasks.

Patient class	Diagnostic tasks		Early detection tasks			
	Binary	Ternary	12 months	24 months	36 months	48 months
CN	542	542	1224	963	659	549
MCI	-	1113				
AD	411	411	106	196	197	144
Total	953	2066	1340	1159	856	693

AD) and ternary (CN vs. AD vs. MCI) classifications, reflecting real-world clinical differentiation challenges. Additionally, EMCI and LMCI patients were grouped into a single MCI category, as the primary focus of this study was to differentiate MCI from CN and AD, rather than analyzing MCI subgroups individually.

The early detection tasks focused on predicting disease progression by classifying AD vs. CN+MCI and grouping CN and MCI patients into a single class. This classification reflected whether treatment would be required (AD patients) or not (CN+MCI patients) at four future time points (12, 24, 36, and 48 months) from recruitment. The goal was to identify patients at risk of developing AD who could benefit from pharmacological therapy to slow disease progression.

Table 2 shows the distribution of patients across different tasks and time points, highlighting the patient classification at baseline and subsequent classifications relevant to the early detection tasks. For these tasks, baseline groups included CN, EMCI, and LMCI classes, whereas patients diagnosed with AD at baseline were excluded from the early detection analysis due to their non-conversion status.

4. Methods

This section details the stages of our method, IMBALMED, also depicted in Fig. 2 distinguishing between the training and testing stages. For notation, we will use the following conventions: bold lowercase letters for vectors, italics uppercase letters for sets, uppercase letters for matrices, and lowercase letters for scalars. In the training phase (Train block in Fig. 2), data balancing involves creating distinct subsets for each modality, differentiated by their class representativeness. This process creates diverse model experts, where the diversity is driven by varying levels of class representativeness. Next, each balanced subset undergoes training using a classifier. During the testing phase (Test block in Fig. 2), given an input sample $\mathcal{X}$, a set of vectors each characterized by different lengths based on the number of features in each modality, class membership probabilities $\mathbf{p}_{ij}$ are calculated, where i refers to the modality and j to the balanced subset. The unimodal fusion then combines these probability distributions within each modality, producing unimodal probability vectors $\mathbf{p}_i$. Finally, the multimodal fusion averages the unimodal probabilities across modalities, yielding the output vector $\mathbf{p}$, which represents the probability of class membership for the sample based on the knowledge learned from the various modalities.

The presented methodology introduces a novel multimodal ensemble-based approach that manipulates class distributions during the training of classifiers for each modality. This methodological innovation not only aims to enhance the robustness and performance of individual classifiers but also focuses on their effective integration.

4.1. Data balancing

In this study, we address a multimodal classification task with m modalities using an unbalanced dataset. Given a finite set of modalities $\mathcal{M} = \{M_1, M_2, \dots, M_m\}$, where each M_i represents a distinct data modality for $i = 1, \dots, m$, let $\mathcal{Y} = \{y_1, y_2, \dots, y_c\}$ represent the set of c classes with y_k as the kth class for $k = 1, \dots, c$. To mitigate the challenge of data imbalance and promote classifier diversity, we employed an undersampling strategy. This involved creating the set S_i of the balanced subsets of the modality M_i, defined as $S_i = \{S_{i1}, S_{i2}, \dots, S_{is}\}$, where i is the modality index and s is the number of the balanced subset within that modality. By creating the subsets, we aim to introduce diversity to the trained model through varying levels of class balance. For each balanced subset S_{ij}, with $j = 1, \dots, s$, the combination of class representativeness levels is defined by the vector:

$$\mathbf{b_{ij}} = [b^1_{ij}, b^2_{ij}, \dots, b^c_{ij}], \quad \mathbf{b_{ij}} \in \mathbb{R}^c, \quad b^k_{ij} \in \mathbb{R} \tag{1}$$

where each element of the vector $\mathbf{b_{ij}}$ represents the percentage of representativeness of the kth class for the subset S_{ij}, scaled between 0 and 1. To ensure significant representation for each class, we define r as the minimum representativeness number of samples for each class, expressed as a percentage between 0 and 1. This value r is the increment by which the representativeness percentages vary from the specified r to 1, which corresponds to the case where only one class is present in the dataset. Formally, we have: $0 < r < 1 \quad \forall r \in \mathbb{R}$.

Each representativeness level b^k_{ij} must satisfy the following conditions:

i. Representativeness levels: Each representativeness b^k_{ij} is defined by:

$$b^k_{ij} = w^k_{ij} \cdot r, \quad w^k_{ij} \in \mathbb{N}, r \in \mathbb{R} \tag{2}$$

where w^k_{ij} is a coefficient that scales the minimum representativeness level r for class k. The values $w^k_{ij} \cdot r$ are selected from the set R_c, which represents the set of all possible representativeness levels for all classes. For each balanced subset S_{ij}, we define:

$$\mathbf{b_{ij}} = [w^1_{ij}, w^2_{ij}, \dots, w^c_{ij}] \cdot r \in R_c = \left\{ r, 2r, \dots, \left\lfloor \frac{1}{r} - c + 1 \right\rfloor \cdot r \right\} \tag{3}$$

where $\left\lfloor \frac{1}{r} - c + 1 \right\rfloor$ represents the maximum representativeness of the specific class within the balanced subset S_{ij} for multi-class problems with an odd number c of classes. This value is obtained by calculating the complement to 1 after assigning representativeness values to the other $c - 1$ classes.

ii. Sum of Representativeness: For each class k, b^k_{ij} represents the percentage of representativeness for that class. The sum of the representativeness percentages across all classes within the same balanced subset $\mathbf{b_{ij}}$ must equal 1:

$$\sum_{k=1}^{c} b^k_{ij} = \sum_{k=1}^{c} w^k_{ij} \cdot r = 1 \tag{4}$$

A binary problem ($c = 2$) would have $s = 9$ balanced subsets ($c^3 + 1$), whilst a ternary problem ($c = 3$) would consist of $s = 28$ different balanced subsets. To better clarify the methodology used, in the case of a binary problem with r set at 0.1, the computed representativeness levels were: $\mathbf{b_{i1}} = [0.1, 0.9], \mathbf{b_{i2}} = [0.2, 0.8], \dots, \mathbf{b_{i9}} = [0.9, 0.1]$. For a ternary problem with $r = 0.11$, the computed representativeness levels were: $\mathbf{b_{i1}} = [0.11, 0.11, 0.78], \mathbf{b_{i2}} = [0.11, 0.22, 0.67], \dots, \mathbf{b_{i28}} = [0.77, 0.11, 0.12]$. This strategy ensured that each class maintained a minimum representativeness percentage of r and several samples for each class equal to the representativeness level of the class in question for the minimum number of samples among the classes.

4.2. Data fusion levels

Our framework incorporates a two-level fusion process, first at the unimodal level and then across different modalities, thereby leveraging the strengths of diverse data representations. For each modality $M_i \in \mathcal{M}$ for $i = 1, \dots, m$, we train a set of classifiers $\mathcal{F}_i = \{f_{i1}, f_{i2}, \dots, f_{is}\}$, where $j = 1, \dots, s$ and f_{ij} represents the model trained on the balanced subset S_{ij} of the ith modality. Each classifier f_{ij} maps from the feature space to the class probability space: $f_{ij} : \mathbb{R}^{a_i} \to \mathbb{R}^c$, where a_i denotes the number of features for the ith modality. For a single input sample

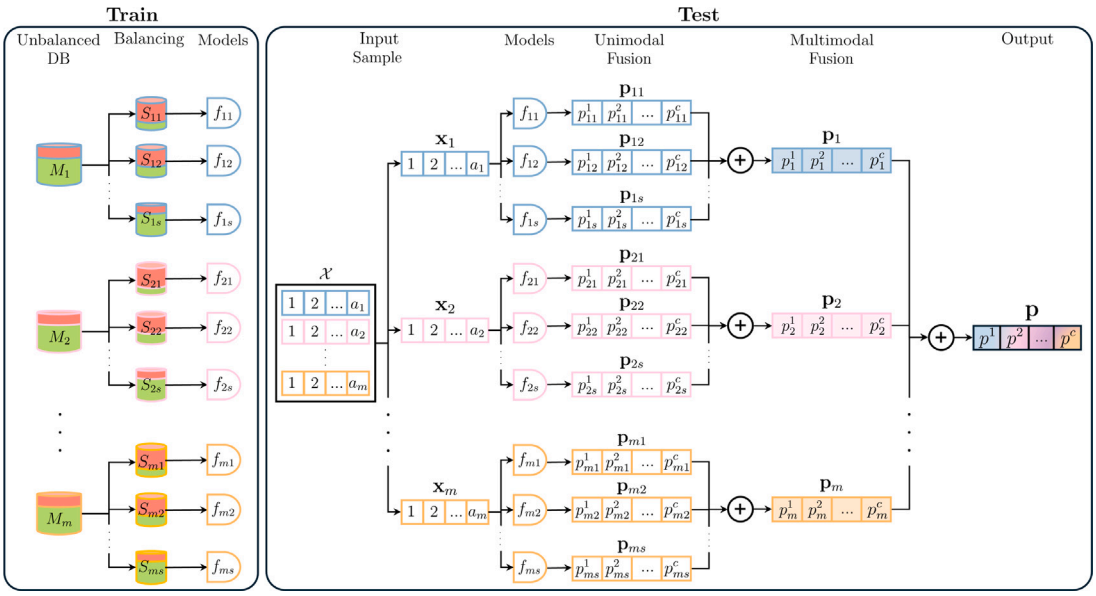


Fig. 2. Schematic representation of the proposed method, highlighting two main steps: training, which includes data balancing and classifier training, and testing, which involves unimodal and multimodal fusion to determine class membership probabilities. The class imbalance is visually represented with red for the minority class and green for the majority class. During data balancing, the sub-datasets are characterized by different class distributions. Different modalities are represented with blue, pink, and orange, indicated by the borders of datasets, sub-datasets, classifiers, and probability arrays. For unimodal fusion, probabilities are displayed as white-filled arrays with border colors corresponding to their modality. In multimodal fusion, arrays are filled with the color of the modality, signifying the integration of information across sub-datasets. The pipeline output is depicted with a black border and multicolored filling, symbolizing the fusion of knowledge from all modalities. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

$\mathcal{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\}$, the $\mathbf{x}_i \in \mathbb{R}^{a_i}$ represents the feature vector for the ith modality in input to the model. The probability that the input sample belongs to a specific class is expressed by the vector $\mathbf{p}_{ij} = f_{ij}(\mathbf{x}_i)$ where $\mathbf{p}_{ij} \in \mathbb{R}^c$ for $i = 1, \dots, m$ and $j = 1, \dots, s$. The vector $\mathbf{p}_{ij}$ is composed of a probability value for each class and expresses the probability that the input sample belongs to each class for the balanced subset S_{ij}, given the classifier f_{ij}.

For the first fusion level, we computed the average of the summed probability distributions $\mathbf{p}_{ij}$ produced by the classifiers:

$$\mathbf{p}_i = \frac{1}{s} \sum_{j=1}^{s} \mathbf{p}_{ij} \tag{5}$$

where each element of $\mathbf{p}_{ij}$ is summed with corresponding elements across all j. The resulting vector $\mathbf{p}_i$ is composed of a probability value for each class and expresses the probability that the input sample belongs to each class for the ith modality.

At the second fusion level, we aggregate the $\mathbf{p}_i$ probability distributions from each modality:

$$\mathbf{p} = \frac{1}{m} \sum_{i=1}^{m} \mathbf{p}_i \tag{6}$$

where each element of $\mathbf{p}_i$ is summed with the corresponding elements across all i. The resulting vector $\mathbf{p}$ contains a probability value for each class, representing the likelihood that the input sample belongs to the specific class, considering all m modalities.

To obtain the final class prediction for the input $\mathcal{X}$, we apply the argmax function to the aggregated probability distribution $\mathbf{p}$, selecting the class with the highest probability:

$$\hat{c}(\mathcal{X}) = \arg\max_{k \in C} \mathbf{p} \tag{7}$$

The predicted class $\hat{c}(\mathcal{X})$ is then used during evaluation to compare with the true class label, assessing the model's performance.

5. Experimental setup

Our study encompasses both diagnostic and early detection tasks. For the diagnostic tasks, we use $m = 4$ data modalities (Section 4.1), excluding neuropsychological test scores to avoid bias, as AD diagnosis often relies on such tests (Weller and Budson, 2018). For the early detection tasks, baseline neuropsychological test scores are included to assess treatment necessity.

To address data imbalance, we apply the IMBALMED methodology described in Section 4.1, setting the minimum representativeness number of samples to $r = 0.1$ for the binary tasks and $r = 0.11$ for the ternary task. These parameters were selected to ensure maximum diversity among the subsets, guaranteeing equal class representation, with $\mathbf{b_{i5}} = [0.5, 0.5]$ for the binary tasks and $\mathbf{b_{i16}} = [0.33, 0.33, 0.34]$ for the ternary task (see Section 4.1 for the notation).

5.1. Training and evaluation

Our training methodology involved a 10-fold cross-validation with $k = 10$ and we evaluated our methodology on 25 different classifiers selected from eight families of learning paradigms. This approach was chosen to demonstrate the robustness and performance of our method across various classification tasks, regardless of the model used. The classifiers were used with default hyperparameters since their tuning is out of the scope of this work. However, the No Free Lunch Theorem states that, in many cases, tuned parameters do not significantly outperform the default ones (Arcuri and Fraser, 2013). The choice of diverse families of learning paradigms aimed to assess the performance across a wide spectrum of classification techniques. Furthermore, testing our method across various learning paradigm families demonstrates its applicability to experts across different domains, not limited to shallow learning models. Table 3 lists the names of the classifiers used in this study, divided by families.

For evaluation, we used the G-mean, which ensures both classes are adequately represented and prevents the model from favoring the majority class, making it highly suitable for evaluating models on imbalanced datasets. For binary tasks, G-mean was calculated as the square root of the product of recall and specificity, whilst for ternary tasks, it was calculated as the cube root of the product of the recall and specificity for each class.

Table 3
Selected classifiers divided by families.

Family	Classifiers
Artificial Neural Networks	Multilayer Perceptron (MLP).
Decision Trees	Decision Tree (DT), Extra-Tree (ET), and Extra-Trees (ETs).
Ensemble Methods	AdaBoost (AB), Gradient Boosting (GB), Hist Gradient Boosting (HGB), Random Forest (RF), XGBoost (XGB), and XGBoost Random Forest (XGBRF).
Label Propagation	Label Propagation (LP).
Label Spreading	Label Spreading (LS).
Linear Models	Linear Discriminant Analysis (LDA), Linear Support Vector (LSVC), Logistic Regression (LR), Passive-Aggressive (PA), Perceptron (Percep.), Ridge (Ridge), Stochastic Gradient Descent (SGD), Support Vector Classification (SVC) and Nu-Support Vector Classification (NuSVC).
Naive Bayes	Bernoulli Naive Bayes (BNB) and Gaussian Naive Bayes (GNB).
Nearest Neighbors	k-Nearest Neighbors (k-NN) and Nearest Centroid (NC).

Table 4
Selected competitors.

Algorithm	Description
Adaptive Random Forest (ARF) (Gomes et al., 2017)	It utilizes the Hoeffding Tree as its base classifier and employs mechanisms to adapt to changes in data distribution over time, known as concept drift.
Adaptative Random Forest with Resampling (ARFR) (Ferreira et al., 2019)	It is an extension of the ARF that incorporates resampling techniques to handle imbalanced data streams.
Comprehensive active learning method for multiclass imbalanced streaming data with concept drift (CALMID) (Liu et al., 2021)	It combines active learning with strategies to address multiclass imbalance and concept drift in data streams, focusing on efficiently selecting the most informative samples for learning.
Leveraging Bagging Adwin (LB) (Bifet et al., 2010)	It leverages the Bagging algorithm with the ADaptive WINdowing (ADWIN (Bifet and Gavalda, 2007)) change detector to handle data with concept drift.
Online AdaBoost (OADA) (Wang and Pineau, 2016)	It is an online version of the AdaBoost algorithm; it updates the model with new data, adjusting weights of incorrectly classified instances to improve performance.
Online Bagging with Adwin (OBA) (Bifet et al., 2009)	It combines the OB algorithm with the ADWIN algorithm for detecting and adapting to concept drift.
Oversampling Online Bagging (OOB) (Wang et al., 2016a)	It is an adaptation of the Online Bagging algorithm (OB (Oza and Russell, 2001)) that incorporates oversampling techniques, by focusing on increasing the representation of minority classes.
Robust Online Self-Adjusting Ensemble (ROSE) (Cano and Krawczyk, 2022)	It dynamically adjusts its composition and parameters in response to concept drift.
Undersampling Online Bagging (UOB) (Wang et al., 2016b)	It extends the OB algorithm by incorporating undersampling techniques, reducing the influence of majority class samples to balance the training process.

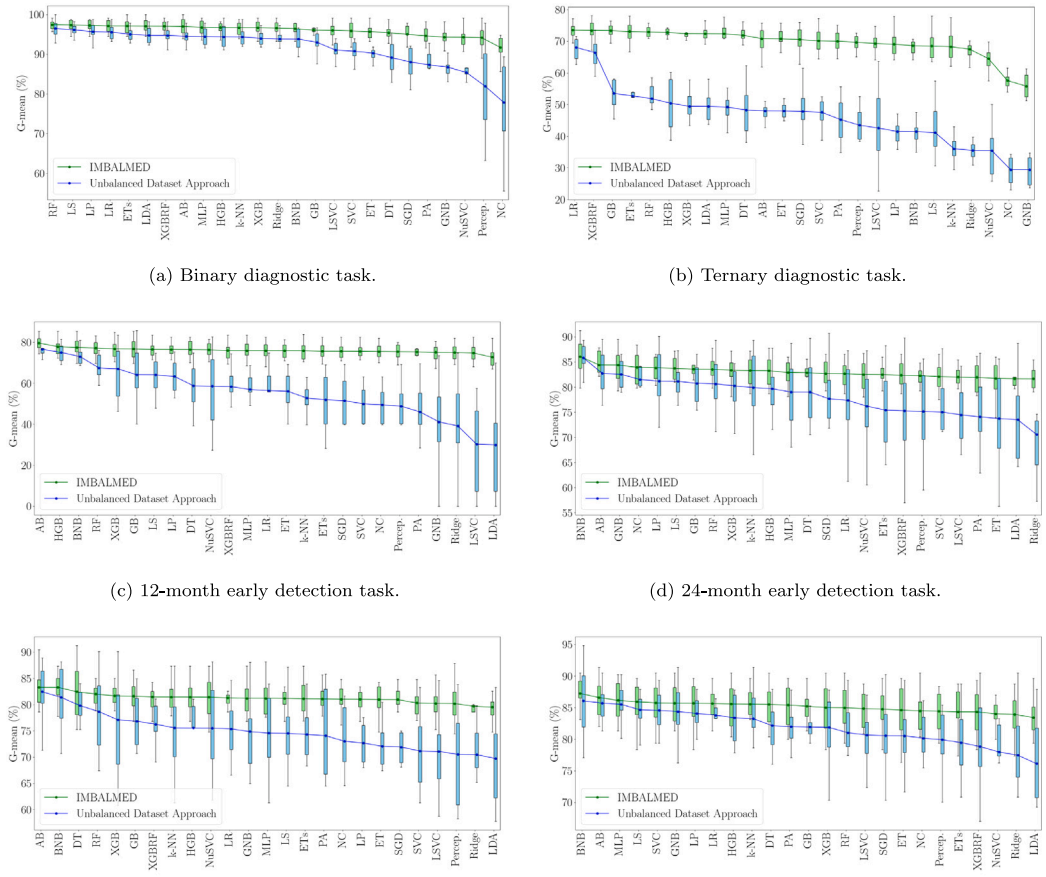
5.2. Competitors and statistical analysis

We compared our balancing method, IMBALMED, with two main approaches: using the raw, unbalanced dataset without any balancing (unbalanced dataset approach), and employing state-of-the-art data balancing algorithms. From a comprehensive review assessing cutting-edge algorithms (Aguiar et al., 2023), we selected the top-performing ensemble-based algorithms for multiclass scenarios, listed in Table 4. To ensure a rigorous comparison, we evaluated the performance on the test set of these algorithms against our most effective classifier, which was selected based on performance on an independent validation set. Each algorithm was implemented with its default classifier and parameters, to adopt the same setup described in 5.1. Furthermore, we assessed the statistical significance of our method IMBALMED against both the unbalanced dataset approach and state-of-art balancing data competitors using the Paired T-test (p-value ≤ 0.05). Given the cross-validation performance values on the test set, we calculated the win rate, representing the instances where IMBALMED's p-value outperformed the specific competitor. To provide a comprehensive view of IMBALMED's effectiveness and consistency, we also measured the tie rate and loss rate, indicating the percentage of times that our method was equal to or worse than the competitors, respectively.

As mentioned in Section 2, there are three related studies addressing imbalance and multimodal learning on the ADNI dataset (Sun et al., 2021; Velazquez and Lee, 2022; Brand et al., 2020). Although they could be valid benchmarks to assess our approach, these works were excluded from our comparison due to insufficient details for a fair experiment reproduction. It is also worth noting that we could not validate the proposed approach on external datasets since the literature does not provide publicly available datasets that comply with all modalities selected in this study.

6. Results and discussions

This section presents the evaluation of our proposed multimodal ensemble-based approach, IMBALMED, across the various diagnostic and early detection tasks. We compare the performance of our method



(a) Binary diagnostic task.

(b) Ternary diagnostic task.

(c) 12-month early detection task.

(d) 24-month early detection task.

(e) 36-month early detection task.

(f) 48-month early detection task.

Fig. 3. IMBALMED vs. Unbalanced Dataset Approach for various tasks. Panels (a) and (b) represent the binary and ternary diagnostic tasks, respectively. Panels (c) to (f) represent the 12-, 24-, 36-, and 48-month early detection tasks, respectively. Lines connect classifiers' average performance.

Table 5
Data fusion performance: top-performing classifier ("Best G-mean" of the "Best Classifier") and average performance across classifiers ("Average G-mean") based on G-mean for each task.

Task	Best G-mean	Best Classifier	Average G-mean
Binary diagnostic	97.40 ± 1.24	RF	96.01 ± 1.33
Ternary diagnostic	73.49 ± 4.31	LR	69.56 ± 4.38
12-month early detection	79.60 ± 3.45	AB	76.10 ± 1.22
24-month early detection	86.10 ± 3.60	BNB	83.01 ± 1.03
36-month early detection	83.30 ± 3.47	AB	81.24 ± 0.88
48-month early detection	87.24 ± 2.76	BNB	85.17 ± 0.86

against both an unbalanced dataset approach and state-of-the-art algorithms designed for handling imbalanced data. The following subsections provide detailed analyses of the performance outcomes, including comparative statistical evaluations.

6.1. Multimodal fusion

Table 5 summarizes the performance of our approach across all downstream tasks. It highlights the "Best G-mean" achieved by the top-performing classifier on the test set (specified in the column "Best Classifier") in each experiment, with the associated standard deviation indicating variability across the cross-validation folds. Additionally, the table presents the "Average G-mean", calculated as the mean G-mean performance across the cross-validation folds and averaged over all classifiers, along with the corresponding standard deviation, reflecting variability in the overall performance of the selected classifiers.

In the diagnostic task, the binary task achieves the greatest robustness in performance, with a low standard deviation of 1.24 in the average G-mean value, whilst the ternary task exhibits greater performance variation, reflecting the increased difficulty of this classification task. In the early detection tasks, performance varied slightly across different time horizons. Both the highest performance and the lowest standard deviation are achieved in the 48-month task. This is because, at this time point, most of the MCI patients have converted to AD, making the task easier. These results highlight the robustness and adaptability of the proposed framework in handling different types of tasks and time horizons. The low standard deviations in average G-mean values across all tasks, especially in the 36-month and 48-month early detection tasks, further emphasize the stability and reliability of the approach. Notably, three out of the six tasks are best performed by ensemble methods, highlighting the power of leveraging a combination of multiple classifiers also known as base learners. This finding highlights their ability to enhance generalizability in complex scenarios and the power of our multimodal ensemble-based approach, which utilizes ensemble techniques as a key component in our framework.

From the unimodal fusion results, as reported in Table B.13 of the Appendix B, it is evident that the Assessment modality performs most effectively across all tasks. In diagnostic tasks, the performance difference between the Assessment modality and the second most effective modality (Image Analysis) is approximately 11.6% for the binary task and 34.8% for the ternary task. In early detection tasks, the performance difference between the two modalities is 4.98%, 2.31%, 3.37%, and 1.97% for the 12-, 24-, 36-, and 48-month tasks, respectively. The smaller performance difference between Assessment and other modalities in early detection tasks is due to the increased complexity of

the task, benefiting more from multimodal data. The smallest difference is observed in the clinically most complex 48-month task: this extended prediction period introduces greater uncertainty and variability, making the task more challenging compared to the 12-, 24-, and 36-month tasks.

6.2. Competitors and statistical analysis

The performance obtained with our methodology was first compared with the performance achieved using the unbalanced dataset approach. Fig. 3 displays the results of these comparisons for the diagnostic tasks (Figs. 3(a) and 3(b)) and the early detection tasks at 12-month (Fig. 3(c)), 24-month (Fig. 3(d)), 36-month (Fig. 3(e)) and 48-month (Fig. 3(f)). Each figure contains 25 boxplots per method, corresponding to the 25 selected classifiers. The green boxplots represent the performance achieved with our balancing method IMBALMED across the folds of the cross-validation, whilst the blue boxplots depict the performance from the unbalanced approach, also across the 10 folds. A green and blue line connects the average performance for each classifier from the balanced and unbalanced methods, respectively. The boxplots are organized in descending order, starting with the classifier that achieved the highest average G-mean across the 10 folds for the balanced dataset method. Across all conducted comparisons, our method consistently demonstrated higher average performance than the unbalanced approach. Notably, the 12-month early detection task (Fig. 3(c)) showed the greatest difference in performance between the balanced and unbalanced methods, indicating a significant improvement. This is attributed to the more pronounced imbalance in this task's dataset, where the majority class CN+MCI constitutes 91.34% of the recruited patients.

In our comparative analysis against nine cutting-edge algorithms for imbalanced data, Table 6 shows the average G-mean across the cross-validation folds for each algorithm. As previously mentioned in Section 5.2, for this comparative analysis we evaluated the performance of state-of-the-art algorithms against the best-performing classifier with our proposed balancing method IMBALMED, which was selected based on its performance on an independent validation set. Our top-performing classifiers in the validation set were: AB for the binary diagnostic task, LDA for the ternary diagnostic, LS for the 12-month early detection task, LR for the 24- and 48-month early detection tasks, and HGB for the 36-month early detection task. Their performances on the test set are reported in the IMBALMED rows for each task. The W-T-L columns in Table 6 report the win, tie, and loss rates, respectively. For each competitor, these columns also include a notation in brackets: W if the win rate exceeds both the tie and loss rates; L if the loss rate dominates; X when the win rate and loss rate are equal. We highlight in bold the tasks where our performance demonstrates a higher win rate against more than half of the competitors. Tasks where IMBALMED's performance is superior to the competitors are shaded with a gray background. Notably, experiments indicating statistical significance with win/tie/loss rates are highlighted with a * symbol.

For the binary diagnostic task (Table 6), our method showed superior performance in all nine experiments, with statistically significant win rates across the board. In the ternary task (Table 6), our balancing method outperformed all competitors, with seven out of nine of these having statistically significant win rates that exceeded both tie and loss rates. Both binary and ternary tasks had no experiments where the loss rate was higher than both win and tie rates. Notably, in the binary task, our method showed the lowest standard deviation, highlighting the robustness of our method. Although in the ternary task, our standard deviation was not the lowest, we still achieved a 1.34% performance improvement over the top competitor, the ARFR algorithm.

In the 12-month early detection task (Table 6), IMBALMED surpassed all competitors, with a win rate higher than both tie and loss rates, and eight of these showing statistical significance. In this 12-month task, IMBALMED had the highest performance, improving by 1.92% over the best competitors CALMID and LB. For the 24-month early detection task (Table 6), our method outperformed eight out of nine competitors, with seven of these showing higher win rates than tie and loss rates, and four of these had statistical significance. The ARFR classifier achieved the highest performance, improving ours by 0.35%; however, it is notable that the standard deviation of our method is 39.44% lower than that obtained by the ARFR algorithm. Additionally, in comparison with the OOB algorithm, our method demonstrates a loss rate higher than both the win rate and the tie rate. This indicates that in more than half of the 10-folds, OOB's performance is superior, but the overall average across the 10-folds is higher with our method, allowing a performance increase of 0.80% with IMBALMED and a standard deviation 45.22% lower. In the 36-month task (Table 6), IMBALMED was superior to all competitors, with seven of them having a higher win rate than both tie and loss rates, and six of these showing statistical significance. It is noteworthy that the OBA algorithm demonstrates equal win and loss rates, with a performance 1.75% lower than ours. Additionally, the OOB algorithm shows a higher win rate compared to our method, but with a performance of 1.99% lower and a standard deviation 55.83% higher, underscoring the robustness of our method. In the 48-month early detection task (Table 6), our method excelled across all competitors, with every experiment showing a higher win rate than both tie and loss rates, four of which had statistical significance. In this 48-month task, IMBALMED had the highest performance, improving by 1.64% over the best competitor (OBA). It is important to emphasize that in the clinically significant 48-month task, our IMBALMED method achieved the highest performance compared to other competitors. The 48-month early detection task is particularly crucial because it allows for a longer-term forecast of AD progression, providing critical insights for early intervention and treatment planning. Shorter-term predictions at 12, 24, or 36 months, while useful, do not offer the same strategic advantage.

Furthermore, we observed statistically significant win rates in 100% of binary diagnostic tasks, 77.78% of ternary diagnostic tasks, and 88.89%, 44.44%, 66.67%, and 44.44% of 12-, 24-, 36-, and 48-month tasks, respectively, all without any statistically significant loss rates.

Overall, IMBALMED demonstrated superior performance in most tasks and time frames. By addressing class imbalance and enhancing model robustness, IMBALMED offers a powerful tool for clinicians to forecast disease progression, enabling timely interventions. The performance of IMBALMED, especially in the critical 48-month early detection task, underscores its potential to significantly advance the field of AD early detection and diagnosis.

6.3. External validation

To evaluate the generalizability of our proposed approach to real-world scenarios, we conducted an external validation experiment using data from the most recent phase of the ADNI dataset, ADNI 4. External validation was conducted at the conclusion of the primary experiments, approximately eighteen months after the acquisition of data for the primary dataset, by which time the ADNI 4 data had become available. This delay allowed us to leverage the most recent ADNI phase to evaluate the generalizability of our approach in real-world scenarios. It is important to note that no datasets other than ADNI were used, as, to the best of our knowledge, no other comprehensive datasets with this type of feature set and characterization of AD are available. By considering ADNI 4, as an extraction from the ADNI dataset that was not previously selected for our primary experiments, we effectively treated the external validation dataset as distinct from the primary dataset. Additionally, we verified that no patients included in the external validation dataset had been selected for the primary dataset, ensuring a strict separation between the two datasets. For the training phase, we performed 10 random extractions from the original dataset to mirror the 10-fold cross-validation setup used in the initial experiments. These extractions were then processed using the same preprocessing

Table 6
Balancing data competitors on diagnostic (table a) and early detection tasks (tables b and c): W for win rates exceeding tie and loss rates, L for highest loss rates, X for equal win and loss rates, and * for statistically significant outcomes. Tasks with a win rate above half of the competitors are highlighted in bold, while our superior performances are shaded gray.

		Diagnostic tasks			
		Binary		Ternary	
		W-T-L	G-mean	W-T-L	G-mean
Algorithm	IMBALMED	-	**96.98 ± 2.00**	-	**72.36 ± 3.90**
	ARF	90-0-10 (W*)	95.19 ± 2.15	60-0-40 (W)	70.99 ± 5.05
	ARFR	90-0-10 (W*)	95.23 ± 2.12	70-0-30 (W)	71.40 ± 3.94
	CALMID	70-0-30 (W*)	95.27 ± 2.41	100-0-0 (W*)	64.68 ± 3.47
	LB	80-0-20 (W*)	95.31 ± 2.29	90-0-10 (W*)	67.40 ± 4.55
	OADA	80-0-20 (W*)	95.01 ± 2.24	100-0-0 (W*)	65.59 ± 4.35
	OBA	90-0-10 (W*)	94.80 ± 2.76	100-0-0 (W*)	66.89 ± 2.58
	OOB	90-0-10 (W*)	94.71 ± 3.07	100-0-0 (W*)	65.54 ± 2.87
	ROSE	80-0-20 (W*)	95.31 ± 2.29	80-0-20 (W*)	69.93 ± 4.74
	UOB	100-0-0 (W*)	94.02 ± 2.69	100-0-0 (W*)	66.64 ± 3.10

(a) Diagnostic tasks: binary and ternary.

		Early detection tasks			
		12 months		24 months	
		W-T-L	G-mean	W-T-L	G-mean
Algorithm	IMBALMED	-	**76.53 ± 4.18**	-	**82.68 ± 2.58**
	ARF	90-0-10 (W*)	49.42 ± 13.86	70-0-30 (W*)	76.19 ± 8.17
	ARFR	80-0-20 (W*)	65.75 ± 12.62	30-0-70 (L)	82.97 ± 4.26
	CALMID	100-0-0 (W*)	75.09 ± 7.75	70-0-30 (W)	71.46 ± 24.44
	LB	100-0-0 (W*)	75.09 ± 7.75	70-0-30 (W*)	74.68 ± 9.54
	OADA	90-0-10 (W*)	68.49 ± 7.36	60-0-40 (W*)	78.73 ± 5.08
	OBA	100-0-0 (W*)	11.52 ± 23.06	70-0-30 (W)	72.78 ± 24.69
	OOB	70-0-30 (W*)	71.35 ± 6.87	40-0-60 (L)	82.02 ± 4.71
	ROSE	80-0-20 (W*)	60.04 ± 15.96	80-0-20 (W*)	76.08 ± 10.27
	UOB	60-0-40 (W)	74.57 ± 4.00	60-0-40 (W)	81.32 ± 5.31

(b) Early detection tasks: 12 and 24 months.

		Early detection tasks			
		36 months		48 months	
		W-T-L	G-mean	W-T-L	G-mean
Algorithm	IMBALMED	-	**81.44 ± 3.07**	-	**85.66 ± 2.43**
	ARF	100-0-0 (W*)	72.40 ± 8.18	90-0-10 (W*)	81.07 ± 3.29
	ARFR	60-0-40 (W*)	75.36 ± 8.36	80-0-20 (W)	84.19 ± 3.99
	CALMID	80-0-20 (W*)	64.57 ± 25.23	60-0-40 (W)	83.10 ± 6.24
	LB	100-0-0 (W*)	65.76 ± 12.01	60-0-40 (W)	83.10 ± 6.24
	OADA	60-0-40 (W*)	78.14 ± 6.17	90-0-10 (W*)	81.83 ± 4.61
	OBA	50-0-50 (X)	80.04 ± 6.74	60-0-40 (W)	84.28 ± 4.28
	OOB	40-0-60 (L)	79.85 ± 6.95	60-0-40 (W)	83.56 ± 5.16
	ROSE	80-0-20 (W*)	77.12 ± 6.35	90-0-10 (W*)	80.26 ± 4.18
	UOB	70-10-20 (W)	78.17 ± 6.74	70-0-30 (W*)	82.22 ± 4.51

(c) Early detection tasks: 36 and 48 months.

Table 7
Distribution of external validation patients for the diagnostic and early detection tasks.

Patient class	Diagnostic tasks		Early detection task
	Binary	Ternary	12 months
CN	344	343	93
MCI	-	65	
AD	28	22	2
Total	372	430	95

pipeline and methodology applied in our primary experiments to ensure consistency in the evaluation framework.

ADNI 4, which began enrolling participants in 2023, lacks data for the 24, 36, and 48 months early-detection tasks, due to the limited follow-up time of patients. Consequently, the external validation was restricted to the 12-month early-detection task and the diagnostic tasks, both binary and ternary. Participants from ADNI 4 were selected following the criteria outlined in Section 3.2, and the distribution of selected patients is presented in Table 7.

Since not all study files used for feature extraction in the earlier ADNI phases are available in ADNI 4, certain data modalities were incomplete in the external validation. Specifically, Biospecimen and

Table 8
Summary of ADNI data modalities selected for external validation at baseline, detailing the number of features and percentage of missing data for diagnostic and early detection tasks.

Modality	# of features		% of missing data	
	Diagnostic tasks	Early detection tasks	Diagnostic tasks	Early detection tasks
Assessment	16	56	0.03	10.70
Subject Characteristics	6	6	0.52	0

Table 9
External validation data fusion performance: top-performing classifier ("Best G-mean" of the "Best Classifier") and average performance across classifiers ("Average G-mean") based on G-mean for each task.

Task	Best G-mean	Best Classifier	Average G-mean
Binary diagnostic	83.04 ± 3.50	GB	77.62 ± 4.28
Ternary diagnostic	49.90 ± 5.83	ET	43.21 ± 5.40
12-month early detection	96.88 ± 0.59	k-NN	83.71 ± 16.85

Image Analysis modalities were excluded due to the absence of the *ADNIMERGE* study file, which is only available for ADNI 1, GO, 2, and 3. Table 8 provides a summary of the selected modalities for external validation, detailing the number of features per task and the percentage of missing data. Furthermore, a detailed list of selected features for each modality and task is provided in Table A.12 in Appendix A.

The results of the multimodal fusion, trained on the dataset described in Section 3, are summarized in Table 9. As outlined in Section 6.1, this table reports the performance across binary and ternary diagnostic tasks and the 12-month early-diagnostic task. Performance metrics include the "Best G-mean", representing the highest-performing classifier on the external validation test set, and the "Average G-mean", which averages the G-mean performance across cross-validation folds and classifiers. Compared to the results presented in Table 5, external validation revealed performance reductions of 20.3% for the binary diagnostic task and 41.2% for the ternary diagnostic task. This decline may be attributed to differences in patient characterization within the ADNI 4 cohort. Notably, ADNI 4 does not provide a study file distinguishing between EMCI and LMCI, despite these subgroups representing distinct pre-AD stages. LMCI patients are generally easier to differentiate from CN individuals, just as EMCI patients are more distinct from AD cases. Conversely, LMCI and AD patients share several overlapping patterns, making their differentiation more challenging. The lack of this distinction could have impacted classification performance, and we hypothesize that, compared to patients in the primary dataset, those in the external validation dataset may exhibit different clinical characteristics, affecting model performance. In contrast, as shown in Table 9, the early detection task demonstrated a 10% performance improvement in the external validation dataset. Several factors could explain this outcome, stemming from a combination of the class-balancing method and the structure of the external validation dataset. Specifically, these results may indicate that the AD class in the early detection task is easier to distinguish compared to the AD class in the primary dataset. However, this hypothesis is refuted by the comparative analysis, which supports a second hypothesis: that IMBALMED is particularly well-suited for handling highly imbalanced datasets, such as the external dataset, where the AD class constitutes only 2.11%.

Our proposed balancing method consistently outperformed the unbalanced dataset approach across all tasks, demonstrating superior performance in external validation metrics. Compared to the average G-mean values of our method shown in Table 9, the unbalanced approach achieved 73.19 ± 5.41, 18.88 ± 14.03, and 49.35 ± 38.86 for the binary diagnostic task, ternary diagnostic task, and early detection task, respectively. The performance trends for each classifier, comparing the two methods, are illustrated in Fig. C.4, which mirrors the results shown in Fig. 3 for the primary dataset. To avoid overloading the presentation of results, Fig. C.4 has been included in Appendix C.

Additionally, the comparative analysis with nine state-of-the-art algorithms for imbalanced data further underscores the efficacy of our approach. These results, detailed in Table 10, compare the performance of competitor algorithms with that of our IMBALMED method, using the top-performing classifiers selected from the validation set of the primary dataset, as described in the previous section on *Competitors and Statistical Analysis*. The results show that in the binary diagnostic task, our method achieves performance comparable to that of competing approaches. However, in the ternary task, IMBALMED outperforms 7 out of 9 competitors; furthermore, most competing methods exhibit a G-mean well below 40%, highlighting their weakness in this type of classification task. This performance gap is likely due to differences in the distribution of patients between the ternary task dataset and the primary dataset. Specifically, in the primary dataset, MCI patients accounted for 53.9% of the ternary classification task, whereas in the external validation dataset, MCI patients represent only 15.1%. This stark difference suggests that, unlike IMBALMED, competing methods struggle to generalize effectively across different datasets. In the early detection task, the comparative analysis highlights that IMBALMED outperforms the competitor methods significantly. Notably, 3 out of 9 competitor methods achieved a G-mean of 0, indicating a complete inability to distinguish the classes correctly in the external validation dataset. This suggests that the patterns associated with AD patients in this dataset are not easily recognizable by conventional methods. However, the diversity strategy adopted by IMBALMED proves crucial in a scenario with extreme class imbalance, improving generalization capabilities and capturing the subtle features of the minority class. These results emphasize the effectiveness of IMBALMED in integrating diversity through class balancing to handle highly complex and imbalanced cases, such as those encountered in real-world clinical practice.

6.4. IMBALMED as a CDSS in clinical practice

The proposed IMBALMED framework has significant potential to function as a Clinical Decision Support System (CDSS) in the diagnosis and progression monitoring of AD. By leveraging its multimodal ensemble approach, IMBALMED effectively addresses the challenges of data imbalance often encountered in real-world clinical datasets. As a CDSS, it would augment medical decision-making by providing clinicians with robust, data-driven insights derived from heterogeneous patient data, such as imaging, cognitive scores, and clinical records. IMBALMED's integration into clinical workflows could be seamlessly achieved by embedding it within existing EHR systems, enabling real-time analysis of patient data. This would allow clinicians to access model predictions and associated confidence scores directly during consultations, facilitating informed and timely decision-making. Notably, IMBALMED is designed to take as input the same diagnostic exams already performed in routine clinical practice, ensuring no additional or invasive testing is required for its operation. This feature positions IMBALMED as both a double confirmation tool for neurologists—offering a second, data-driven opinion during diagnosis—and an early warning system for potential disease progression. During patient visits, the neurologist would benefit from dual insights: the model's perspective on the patient's current condition and its prediction of future disease trajectory. This dual perspective, derived from the same standard diagnostic data, supports more comprehensive and confident clinical decision-making. Furthermore, the deployment of IMBALMED as a CDSS must adhere to stringent regulatory and ethical standards to ensure patient safety and data privacy. Clinical validation studies should be conducted

Table 10
Balancing data competitors on diagnostic and early detection tasks for the external validation: W for win rates exceeding tie and loss rates, L for highest loss rates, X for equal win and loss rates, and * for statistically significant outcomes. Tasks with a win rate above half of the competitors are highlighted in bold, while our superior performances are shaded gray.

		Diagnostic tasks				Early detection task	
		Binary		Ternary		12 months	
		W-T-L	G-mean	W-T-L	G-mean	W-T-L	G-mean
Algorithm	IMBALMED	-	81.89 ± 4.07	-	**46.63 ± 9.20**	-	**96.22 ± 0.67**
	ARF	30-0-70 (L)	83.43 ± 2.47	100-0-0 (W*)	17.29 ± 17.73	90-0-10 (W*)	37.85 ± 38.69
	ARFR	40-10-50 (L)	83.23 ± 2.53	80-0-20 (W*)	33.17 ± 22.21	80-20-0 (W*)	77.40 ± 28.46
	CALMID	60-0-40 (W)	82.40 ± 1.73	100-0-0 (W*)	21.68 ± 14.76	100-0-0 (W*)	0.00
	LB	40-0-60 (L)	82.62 ± 3.23	90-0-10 (W*)	25.16 ± 17.97	100-0-0 (W*)	0.00
	OADA	50-0-50 (X)	81.92 ± 2.48	60-0-40 (W)	48.24 ± 2.89	90-10-0 (W*)	67.47 ± 35.64
	OBA	60-0-40 (W)	81.32 ± 0.87	60-0-40 (W)	49.96 ± 2.22	100-0-0 (W*)	0.00
	OOB	60-0-40 (W)	81.38 ± 1.04	100-0-0 (W*)	17.36 ± 15.59	90-10-0 (W)	81.95 ± 28.48
	ROSE	50-0-50 (X)	81.40 ± 2.37	60-0-40 (W)	42.69 ± 15.89	70-20-10 (W*)	56.89 ± 38.93
	UOB	60-0-40 (W)	81.03 ± 0.89	100-0-0 (W*)	19.29 ± 17.73	100-0-0 (W)	82.82 ± 27.66

to demonstrate the model's reliability, transparency, and generalizability across diverse healthcare settings. Once validated, IMBALMED could significantly enhance clinical practice by supporting early diagnosis, guiding personalized treatment plans, and monitoring disease progression, ultimately improving patient outcomes.

7. Conclusions

The rising prevalence of AD, the limited sensitivity of traditional diagnostic methods, and the societal and economic burdens it imposes underline the critical need for advanced diagnostic and early detection methodologies. In this study, we introduced a novel methodology for addressing data imbalance, that leverages the strengths of multimodal ensemble learning while incorporating multi-level data fusion. For each of the four selected tabular modalities from the ADNI database (Anon, 2024), we train a series of classifiers on varied class distributions followed by a fusion strategy that integrates the different modalities. Our framework was evaluated on two diagnostic tasks (binary and ternary) and four binary early detection tasks (at 12, 24, 36, and 48 months) and it was compared with the standard unbalanced dataset approach and with state-of-the-art imbalanced data algorithms. Through our proposed multi-level data fusion strategy, our method IMBALMED has proven superior to unbalanced approaches, enhancing diversity and effectively boosting performance. Comparative analysis with state-of-the-art algorithms revealed that our approach enhances performance in all experiments on binary and ternary diagnostic tasks, 12-, 36-, and 48-month early detection tasks experiments, and eight out of nine experiments on the 24-month early detection task.

To further validate our methodology and assess its real-world generalizability, we conducted an external validation using the most recent phase of the ADNI dataset. The results revealed that our method excelled in the early detection tasks, the most severely imbalanced dataset. IMBALMED emerged as the best-performing model among all competitors, underscoring its effectiveness in scenarios where other traditional methods often struggle due to extreme class imbalance. This highlights IMBALMED's ability to maintain high predictive performance in such conditions, reinforcing its potential for clinical application as a reliable tool for identifying at-risk individuals before significant cognitive decline.

A limitation of our study is that all patients were sourced from a single dataset, the ADNI. Whilst this was acceptable for our research, clinical implementation of our model would benefit from additional datasets, even of different diseases, to capture greater feature variability and to study performance variation. Furthermore, although unimodal analyses have shown that all modalities are nearly equally informative for early detection tasks, the diagnostic task reveals a greater influence of the Assessment modality. Therefore, conducting an ablation study on modalities would be significant, as it could further enhance performance.

CRediT authorship contribution statement

Arianna Francesconi: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lazzaro di Biase:** Supervision, Conceptualization. **Donato Cappetta:** Funding acquisition. **Fabio Rebecchi:** Funding acquisition. **Paolo Soda:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Rosa Sicilia:** Writing – review & editing, Visualization, Validation, Supervision, Software, Project administration, Methodology, Formal analysis, Conceptualization. **Valerio Guarrasi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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tics.

Appendix A. Features selected

See Tables A.11 and A.12.

Appendix B. Unimodal fusion performance

See Table B.13.

Appendix C. External validation: IMBALMED vs. Unbalanced dataset approach

Fig. C.4 presents a comparative analysis between our method and an approach using an unbalanced dataset. The performance trends of the unbalanced method, particularly in the early detection task, it is clear that only 5 out of 25 classifiers approach the performance achieved with IMBALMED, highlighting the importance of balancing in such a highly imbalanced task. In the ternary task, despite generally low performance, only one classifier comes close to IMBALMED's performance, surpassing it by about 2 percentage points.

Table A.11
Primary Dataset: Detailed list of selected features by modality and task.

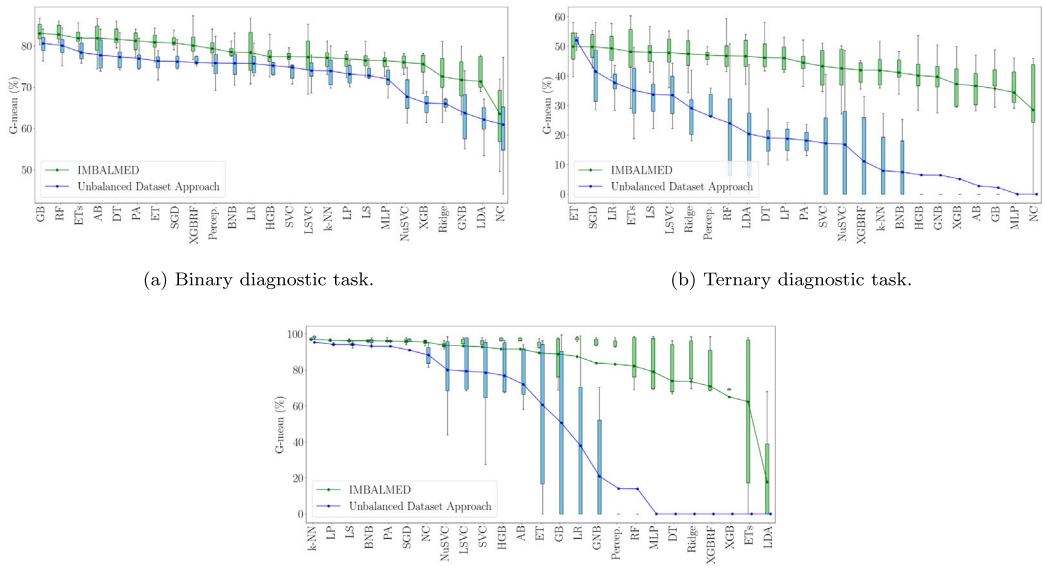
Modality	Diagnostic Tasks	Early Detection Tasks
Assessment	FAQ_bl, FAQSOURCE, FAQFINAN, FAQFORM, FAQSHOP, FAQGAME, FAQBEVG, FAQMEAL, FAQEVENT, FAQTV, FAQREM, FAQTRAVL, HMONSET, HMSTEPWS, HMSOMATC, HMEMOTIO, HMHYPERT, HMSTROKE, HMNEURSM, HMNEURSG, HMSCORE, GDSATIS, GDDROP, GDEMPTY, GDBORED, GDSPIRIT, GDAFRAID, GDHAPPY, GDHELP, GDHOME, GDMEMORY, GDALIVE, GDWORTH, GDENERGY, GDHOPE, GDBETTER, GDTOTAL	FAQ_bl, FAQSOURCE, FAQFINAN, FAQFORM, FAQSHOP, FAQGAME, FAQBEVG, FAQMEAL, FAQEVENT, FAQTV, FAQREM, FAQTRAVL, HMONSET, HMSTEPWS, HMSOMATC, HMEMOTIO, HMHYPERT, HMSTROKE, HMNEURSM, HMNEURSG, HMSCORE, GDSATIS, GDDROP, GDEMPTY, GDBORED, GDSPIRIT, GDAFRAID, GDHAPPY, GDHELP, GDHOME, GDMEMORY, GDALIVE, GDWORTH, GDENERGY, GDHOPE, GDBETTER, GDTOTAL, PHC_Diagnosis, PHC_MEM, PHC_EXF, PHC_LAN, WORD1, WORD2, WORD3, MMWATCH, MMPENCIL, MMREPEAT, MMHAND, MMFOLD, MMONFLR, MMREAD, MMWRITE, MMDRAW, MMSCORE, CLOCKCIRC, CLOCKSYM, CLOCKNUM, CLOCKHAND, CLOCKTIME, CLOCKSCOR, COPYCIRC, COPYSYM, COPYNUM, COPYHAND, COPYTIME, COPYSCOR, AVTOT1, AVERR1, AVTOT2, AVERR2, AVTOT3, AVERR3, AVTOT4, AVERR4, AVTOT5, AVERR5, AVTOT6, AVERR6, AVTOTB, AVERRB, CATANIMSC, CATANPERS, CATANINTR, TRAASCOR, TRAAERRCOM, TRAAERROM, TRABSCOR, TRABERRCOM, TRABERROM, AVDEL30MIN, AVDELERR1, AVDELTOT, AVDELERR2, ADNI_MEM, ADNI_EF, ADNI_LAN, ADNI_VS, ADNI_EF2, FAQ_bl.1, MMSE_bl, CDRSB_bl, RAVLT_immediate_bl, RAVLT_learning_bl, RAVLT_forgetting_bl, RAVLT_perc_forgetting_bl, LDELTOTAL_BL, TRABSCOR_bl, mPACCdigit_bl, mPACCtrailsB_bl, DX_bl
Biospecimen	APOE4	APOE4, TAU_ADNIMERGE, PTAU_ADNIMERGE, DX_bl
Image Analysis	FDG_bl, FDG, Ventricles_bl, Hippocampus_bl, WholeBrain_bl, Entorhinal_bl, Fusiform_bl, MidTemp_bl, ICV_bl	FDG_bl, FDG, Ventricles_bl, Hippocampus_bl, WholeBrain_bl, Entorhinal_bl, Fusiform_bl, MidTemp_bl, ICV_bl, DX_bl
Subject Characteristics	AGE, PTGENDER, PTEDUCAT, PTETHCAT, PTRACCAT, PTMARRY	AGE, PTGENDER, PTEDUCAT, PTETHCAT, PTRACCAT, PTMARRY, DX_bl

Table A.12
External Validation Dataset: Detailed list of selected features by modality and task.

Modality	Diagnostic Tasks	Early Detection Tasks
Assessment	GDSATIS, GDDROP, GDEMPTY, GDBORED, GDSPIRIT, GDAFRAID GDHAPPY, GDHELP, GDHOME, GDMEMORY, GDALIVE, GDWORTH, GDENERGY, GDHOPE, GDBETTER, GDTOTAL	FAQFINAN, FAQFORM, FAQSHOP, FAQGAME, FAQBEVG, FAQMEAL, FAQEVENT, FAQTV, FAQREM, FAQTRAVL, CLOCKCIRC, CLOCKSYM, CLOCKNUM, CLOCKHAND, CLOCKTIME, CLOCKSCOR, COPYCIRC, COPYSYM, COPYNUM, COPYHAND, COPYTIME, COPYSCOR, AVTOT1, AVERR1, AVTOT2, AVERR2, AVTOT3, AVERR3, AVTOT4, AVERR4, AVTOT5, AVERR5, AVTOT6, AVERR6, AVTOTB, AVERRB, CATANIMSC, CATANPERS, CATANINTR, TRAASCOR, TRAAERRCOM, TRAAERROM, TRABSCOR, TRABERRCOM, TRABERROM, BNTSPONT, BNTSTIM, BNTCSTIM, BNTPHON, BNTCPHON, BNTTOTAL, AVDEL30MIN, AVDELERR1, AVDELTOT, AVDELERR2
Subject Characteristics	AGE, PTGENDER, PTEDUCAT, PTETHCAT, PTRACCAT, PTMARRY	AGE, PTGENDER, PTEDUCAT, PTETHCAT, PTRACCAT, PTMARRY

Table B.13
Performance comparison of unimodal modalities for various tasks. The table shows the effectiveness of different modalities in diagnostic and early detection tasks, including performance differences in binary and ternary classification tasks and across different time points.

Modality	Binary Diagnostic	Ternary Diagnostic	12-month Early detection	24-month Early detection	36-month Early detection	48-month Early detection
Assessment	95.42 ± 0.72	71.90 ± 2.40	78.50 ± 3.33	83.71 ± 2.25	81.58 ± 1.61	84.78 ± 3.17
Biospecimen	66.85 ± 4.69	25.98 ± 9.30	71.21 ± 1.42	78.90 ± 1.19	75.15 ± 1.98	81.41 ± 1.25
Image Analysis	85.52 ± 1.29	53.35 ± 2.19	74.77 ± 2.01	81.82 ± 1.11	78.92 ± 1.75	83.14 ± 1.06
Subject Characteristics	59.40 ± 3.47	38.00 ± 5.50	69.41 ± 1.82	78.01 ± 2.31	74.64 ± 2.34	78.65 ± 3.97



(a) Binary diagnostic task.

(b) Ternary diagnostic task.

(c) 12-month early detection task.

Fig. C.4. Panels (a), (b) and (c) represent the binary, ternary diagnostic, and 12-month early detection tasks, respectively. The green boxplots represent the performance achieved by IMBALMED across the folds of the cross-validation, whilst the blue boxplots depict the performance from the unbalanced approach. A green and blue line connects the average performance for each classifier from the balanced and unbalanced methods, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Data availability

The authors do not have permission to share data.

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